



Second Round 2025

– Solutions

Duration: 3 hours

Zürich, Lausanne, Lugano

Difficulty: The problems of each topic are ordered by difficulty.

December 21, 2024

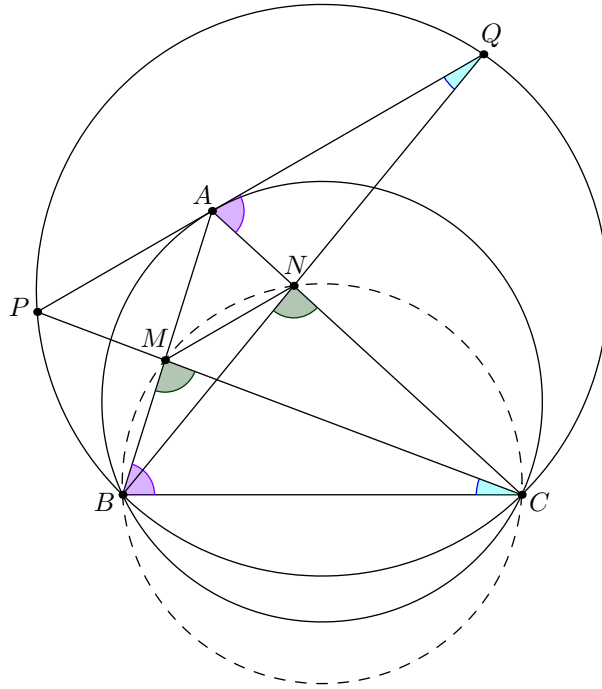
Points: Each problem is worth 7 points.

Preliminary Remark

A complete solution is worth 7 points. For every problem, up to 2 points can be deducted from a full solution for (minor) flaws. Partial marks are attributed according to the marking schemes. In the case of multiple marking schemes for the same problem, the score is the maximum among all the marking schemes.

- G1)** Let ABC be a triangle with circumcircle Ω , and k be a circle passing through B and C such that A is in the interior of k . The tangent to Ω through A intersects k at two points P and Q , such that P and C are on different sides of AB . If M is the intersection of AB and PC and N is the intersection of AC and QB , prove that MN is parallel to PQ .

Solution



Observe that MN is parallel to PQ if and only if angles $\angle MNB$ and $\angle PQB$ are equal. But as $\angle PQB = \angle PCB$ by the inscribed angle theorem in k , it suffices to show that $\angle MNB = \angle MCB$, or in other words that the quadrilateral $BCNM$ is cyclic.

To do so, we let β be the angle at B and δ be the angle $\angle PCB$. On the one hand $\angle CAQ = \beta$ because line QA is tangent to Ω . On the other hand, $\angle AQB = \delta$ by the inscribed angle theorem in k . Therefore,

$$\angle BMC = 180^\circ - \beta - \delta = 180^\circ - \angle CAQ - \angle AQB = 180^\circ - \angle NAQ - \angle AQN = \angle QNA = \angle BNC,$$

where we used that the sum of the interior angles of a triangle is 180° . This proves that $BCNM$ is cyclic, and we can easily conclude by the above argument that MN is indeed parallel to PQ .

Second solution: Here is an alternative way to show that $BCNM$ is cyclic. By introducing γ as the angle at C , we know that

$$\angle NBM = \angle ABQ = \angle PAB - \angle PQB$$

by the exterior angle theorem. But $\angle PAB = \angle ACB = \gamma$ as PA is tangent to Ω , and $\angle PQB = \delta$ as computed above. Therefore, $\angle MBN = \gamma - \delta = \angle ACM = \angle NCM$, and we conclude the same way.

Marking Scheme

First part (5P): Proving that $BCNM$ is cyclic

- 1P: Noting at least one of the following:
 - $\angle QAC = \angle ABC$ or $\angle PAB = \angle ACB$ using the tangent chord theorem
 - $\angle QPC = \angle QBC$ or $\angle PQB = \angle PCB$ using cyclicity of $BCQP$
- 3P: Combining the above to calculate either $\angle PMA$ or $\angle QNA$
(finding the similarity $\triangle PMA \sim \triangle BNC$ or $\triangle QNA \sim \triangle CNB$ also counts)
- 3P: Combining the above to calculate either $\angle ABQ$ or $\angle PCA$
- 4P: Showing that $\angle CNB = \angle CMB$, or that $\angle MCN = \angle MBN$
- 5P: Concluding that $BCNM$ is cyclic

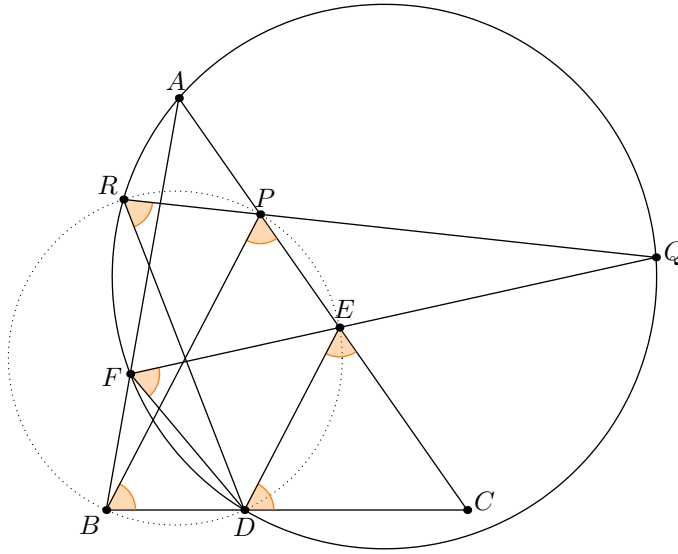
Second part (2P): Concluding

- 1P: Deducing from $BCNM$ cyclic that $\angle MNB = \angle MCB$ or $\angle NMC = \angle NBC$
- 1P: Using that $BCQP$ is cyclic to conclude that $MN \parallel PQ$

The 2 points in the second part can also be awarded if the cyclicity of $BCNM$ has not been proven.

G2) Let ABC be a triangle with $AC > BC$. Its incircle touches sides BC , CA and AB at D , E and F , respectively. Let P be the point on segment AC such that $BP \parallel DE$. Let Ω be the circumcircle of triangle AFD . Line EF intersects Ω again at Q and line PQ intersects Ω again at R . Prove that $PEBR$ is a cyclic quadrilateral.

Solution



Note that $PEDB$ is cyclic, as we have $\angle CDE = \angle CED = \angle CBP = \angle CPB$ from $BP \parallel DE$. It is also the case that $PEDR$ is cyclic, as

$$\angle DRP = \angle DRQ = \angle DFQ = \angle DFE = \angle DEC,$$

where we used the tangent chord theorem in the last equality. As there is a unique circle passing through P , E , and C , these two cyclic quadrilaterals are inscribed in the same circle. In particular, $PEBR$ is cyclic.

Marking Scheme

First part (3P): Proving that $PEDB$ is cyclic

- 1P: claiming that $DC = EC$ or $BC = PC$ (or the equivalent angle equalities)
- 2P: proving an angle equality sufficient to show that $PEDB$ is cyclic, without stating it
- 2P: proving that $PEDB$ is an isosceles trapezoid, without stating that it's cyclic

Second part (3P): Proving that $PEDR$ or $PDBR$ is cyclic

- 2P: proving an angle equality sufficient to show that $PEDR$ or $PDBR$ is cyclic, without stating it

Third part (1P) : Finishing

If no other points are granted, the student is given 1P for claiming that $PEDBR$ is a cyclic pentagon.

- C1) Let n be a positive integer. Pingu the penguin and his n penguin friends are collecting salmon. Every penguin has at most n salmon, and no two penguins have the same number of salmon. In how many ways can the $n + 1$ penguins split into some number of groups of arbitrary sizes, such that each group has exactly n total salmon?

Solution

We prove the following: If n is odd, there are $\frac{n+1}{2}$ possible groupings, such that each group has a collective amount of n salmon. If n is even such an arrangement is not possible, so there are 0 possible groupings.

Note that each penguin has an integer amount of salmon between 0 and n . As there are exactly $n + 1$ such integers and $n + 1$ penguins, there exists a penguin with exactly i salmon for every integer $0 \leq i \leq n$. Let's call this penguin the " i -th penguin", denoted by P_i . Thus, the total amount of salmon is always

$$0 + 1 + 2 + \cdots + n = \frac{(n+1)n}{2}.$$

So if we split this total amount of salmon into k groups each having n salmon, the number of groups k would have to be

$$k = \frac{(n+1)n/2}{n} = \frac{n+1}{2}.$$

Clearly the amount of groups should be an integer, meaning that no such grouping is possible if $n + 1$ is odd i.e. if n is even.

With the case n being even out of the way, assume now that n is odd.

Note that the 0-th penguin could be taken away from his group and assigned to any other group without changing the total amount of salmon in each group. So we will ignore the 0-th penguin for now. We also note that now the n -th penguin will always be alone in its group, as adding any other penguin would strictly increase the amount of salmon above n . So moving forwards, let's also ignore the n -th penguin.

We prove the following claim by induction, starting with $k = n - 1$ down to $k = \frac{n+1}{2}$:

Claim: The group containing the Penguin P_k consist exactly of P_k and P_{n-k} .

Base case, $k = n - 1$: Since P_{n-1} already has $n - 1$ salmons, the only penguin that could still join P_{n-1} without making the amount of salmons of the group greater than n is P_1 .

Induction step: Note that by the induction hypothesis, penguins $P_{n-1}, \dots, P_{k+1}$ and $P_1, \dots, P_{n-k-1}$ are already necessarily assigned to complete groups. Now looking at P_k , the only penguin who isn't yet in a group and doesn't have too many salmons is P_{n-k} . So P_k and P_{n-k} must be in the same group.

Summarizing we have the necessary grouping of the following $\frac{n+1}{2}$ groups:

$$\{P_n\}, \{P_{n-1}, P_1\}, \{P_{n-2}, P_2\}, \dots, \{P_{(n+1)/2}, P_{(n-1)/2}\}.$$

All that is left to do is to count in how many ways we can add the penguin P_0 to the existing grouping. P_0 can't form it's own group, yet can be added to any one of the groups without changing the groups' amount of salmons, resulting in a valid grouping of all penguins. There are exactly $\frac{n+1}{2}$ groups, hence we end up with $\frac{n+1}{2}$ possible groupings after adding P_0 .

Marking Scheme

- (A) **2P:** Prove that in the even case, the answer is 0.
 - (A1) **1P:** State that the total sum of salmons is $\frac{(n+1)n}{2}$.
- (B) **5P:** Prove that in the odd case, the answer is $\frac{n+1}{2}$.
 - (B1) **1P:** Claim that in the odd case, the answer is $\frac{n+1}{2}$. (It must be mentioned that this would only be in the odd case)
 - (B2) **1P:** Explicitly consider the pairing P_k and P_{n-k} , for $k \geq 1$.
 - (B3) **2P:** Prove that (implicitly or not) ignoring P_0 there is only the one grouping. (Note this only works, when n is odd)
 - (B4) **1P:** State that P_0 could be in added/moved to any group.

The marking scheme is additive.

- C2)** The n participants of the Olympiad are trying to escape Wonderland. They arrive at a row of n closed doors, ordered in decreasing size. For every $1 \leq k \leq n$, there is exactly one participant that fits through the first k doors, but not the others. One by one, in some order, the participants approach the row of doors to pass through it. Each participant walks along the row of doors, starting at the largest one. If they come across an open door they fit through, they walk through it and close it behind them. If they reach the last door they fit through and it is closed, they open it and go through it, leaving it open behind them. If at the end all doors are closed again, all the participants can escape safely. In how many different orders can the n participants approach the row of doors while achieving this?

Solution

Answer: For n odd, there are no possibilities. For n even, there are $((n-1)!!)^2$, where (for odd n) the notation $n!!$ denotes $n \cdot (n-2) \cdots 3 \cdot 1$.

Solution:

Firstly, note that the total number of participants has to be even. Every door that is closed at the end must have been passed through by an even number of participants. Let us therefore say the total number of participants is $2n, n \in \mathbb{N}$.

Let $S(2n)$ be the number of ways $2n$ participants can be ordered in such a manner that all the doors are closed at the end. Notice that this value does not change if there are more than $2n$ doors, as long as it remains true that each participant fits through a different number of doors - all that matters is the *relative* ordering of their sizes.

To better see this, call a participant " k " if they fit through the first k doors. Please note that in the original formulation, larger participants fit through *less* doors so this numerical mapping is in fact inverted; 1 is the largest participant and n is the smallest. We can see that the ordering

$$4, 6, 3, 1, 2, 5$$

is valid, because 4 will open the 4th door; 6 will also go through the open 4th door and shut it behind them; 3 will open the 3rd door; 1 will open the 1st door; 2 will go through the open 1st door and shut it behind them, and finally 5 will go through the open 3rd door and shut it behind them. Likewise, if we instead had ten doors and the six participants $\{1, 3, 5, 6, 8, 9\}$ then

$$6, 9, 5, 1, 3, 8$$

is also valid - because the relative ordering is the same as before (4th biggest, 6th biggest, etc.). One could also see this just by pretending the doors corresponding to the participants that aren't present - 2, 4, 7, 10 - don't exist, since they will never be used. By ignoring these doors, we find ourselves in an analogous situation to the first example.

Now let us return to the original problem. Participant 1 will always open the first door and the participant after them will always also pass through the first door and close it behind themselves. If they are the last participant then the first door will remain open, so this is impossible. Consequently, if we remove 1 and the participant immediately after them, the remaining $2n - 2$ participants should still constitute a valid ordering - except that we have proven above that even with $2n$ doors, the number of ways of ordering them is still $S(2n)$.

It follows that $S(2n) = (2n-1)^2 \cdot S(2n-2)$ as we have $2n-1$ choices of where to put 1 - anywhere except at the very end - we have $2n-1$ choices of which participant m will be right after 1,

and after removing 1 and m , we have $S(2n - 2)$ ways of arranging the remaining numbers. Via this recursive formula, and the fact that $S(2) = 1$ (as 1, 2 is a valid ordering but 2, 1 is not) we obtain that

$$S(2n) = (2n - 1)^2 \cdot (2n - 3)^2 \cdots (1)^2 = ((2n - 1)!!)^2 = \left(\frac{(2n)!}{2^n n!} \right)^2$$

and we are done.

Solution 2: (David Reichmuth, Myriam Faltin)

As above, note that n odd is impossible and that we work with $2n$ instead. Note that every door that is used has exactly two participants that pass through it, as it cannot be opened by two different participants, and so is opened by exactly one and closed by exactly one. We put the participants into pairs (X_1, X_2) if they pass through the same door, with X_1 passing first - clearly, X_1 is the larger participant. We say pair $(X_1, X_2) > (Y_1, Y_2)$ if X_1 is larger than Y_1 .

We see that, if $(X_1, X_2) > (Y_1, Y_2)$, then the four participants X_1, X_2, Y_1, Y_2 cannot appear in the following orders (removing everyone else in between):

- X_1, Y_1, X_2, Y_2 : Y_1 would close the door opened by X_1 .
- X_1, Y_1, Y_2, X_2 : Same reason as above.
- Y_1, X_1, Y_2, X_2 : Y_2 would close the door opened by X_1 .

The other three orderings are possible.

Now let us assume we have the n pairs already and see how many ways they can be placed together, just as a function of the ordering on pairs. If we consider the largest pair first, the two elements appear together and so appear in $2n - 1$ places relative to the other $2n - 2$ participants; and by repeating this reasoning we obtain $(2n - 1)!!$ ways of placing the participants. Analogously, we can go from the smallest pair to the largest; in this case we have 1 ways to place the smallest pair; then 3 ways to place the second smallest pair relative to it; then 5 ways of placing the third smallest pair relative to the first two, and so forth. Finally, we note that there are $(2n - 1)!! = \frac{2n!}{n! 2^n}$ ways of creating the pairs (any partition of $2n$ elements into pairs corresponds to a valid creation of pairs, since we are ordering them).

Marking Scheme

Here, $n \in \mathbb{N}$ and $S(n)$ denotes the number of possible ways for n doors and n participants of different sizes, and $T(n)$ denotes the number of possible ways for $n + 2$ doors and n participants of different sizes.

Participants get the maximum of their scores from the two schemes.

Solution 1.

- (A) **1P:** Claim $S(2n) = ((2n - 1)!!)^2$ or equivalent.
- (B) **2P: Generalisation:** Prove that $T(2n - 2) = S(2n - 2)$.
 - (B1) **1P:** Claim that $T(2n - 2) = S(2n - 2)$. Any claim of a more general statement is fine.
 - (B2) **1P:** Any correct justification of the above claim. The justification does not have to be very rigorous; something like *"removing the doors corresponding to the removed participants and relabelling the other doors is the same problem"* or *"considering just the relative sizes of the participants, finding a valid ordering is the same as if there were $2n-2$ doors"* is sufficient.
- (C) **4P: Induction:** Prove that $S(2n) = (2n - 1)^2 \cdot T(2n - 2)$.
 - (C1) **1P:** Remarking that the largest participant always opens the first door and is proceeded by someone who closes it.
 - (C2) **2P:** Noting that if we remove these two then the remaining participants constitute a valid ordering for $2n - 2$ participants, albeit with two additional doors.
 - (C3) **1P:** Showing that there are $(2n - 1)^2$ ways to place the largest participant in a valid position and the size of the proceeding participant.

Solution 2.

- (A) **1P:** Claim $S(2n) = ((2n - 1)!!)^2$ or equivalent.
- (B) **2P: Generalisation:** Considering pairs.
 - (B1) **2P:** Noting that exactly two participants pass through each door and ordering the pairs induced in this way by the larger of the two participants.
- (C) **4P: Induction:** Pair descent.
 - (C1) **1P:** Remarking that the largest pair always consists of two consecutive elements.
 - (C2) **1P:** Noting that there are $2n - 1$ ways to remove this pair.
 - (C3) **2P:** Finishing by remarking there are $\frac{2n!}{n!2^n} = (2n - 1)!!$ ways to pick the pairs.

Considering the pairs in ascending order is analogous to doing (E1), (E2) and a complete justification for why there are $1 \cdot 3 \cdot \dots \cdot 2n - 1$ ways to place the pairs is considered equivalent to these two points.

Failing to state and explain why n cannot be odd is a minor flaw with the removal of 1P **from full solutions**.

N1) Let a, b be positive integers. Prove that the expression

$$\frac{\gcd(a+b, ab)}{\gcd(a, b)}$$

is always a positive integer, and determine all possible values it can take.

Solution

The values attained by the expression are exactly all the positive integers. We first prove that the expression must evaluate to a positive integer. Let a, b be positive integers and write $d = \gcd(a, b)$. Note that since $d \mid a$ and $d \mid b$, we must have $d \mid a+b$ and $d \mid ab$. Thus, $d = \gcd(a, b) \mid \gcd(a+b, ab)$ and the expression is integral, clearly positive.

To see why the expression can take on any arbitrary positive integer value n , consider the positive integers

$$a = n \quad \text{and} \quad b = 2n^2 - n.$$

We have $\gcd(a, b) = n$, as well as

$$\gcd(a+b, ab) = \gcd(2n^2, 2n^3 - n^2) = n^2.$$

Thus, we have

$$\frac{\gcd(a+b, ab)}{\gcd(a, b)} = \frac{n^2}{n} = n.$$

Marking Scheme

- (A) **1P:** Clearly claiming the correct answer or giving constructions for 1, 2, 3.
- (B) **2P:** Proving the expression is integral.
 - (B1) **1P:** Writing $a = dx$ and $b = dy$ for $d = \gcd(a, b)$.
- (C) **4P:** Proving the expression can take on any positive integer value.
 - (C1) **1P:** Writing $a = dx$ and $b = dy$ for $d = \gcd(a, b)$ and reducing the expression.
 - (C2) **1P:** Attempting a construction where $a \mid b$ and reducing the expression.
 - (C3) **1P:** Attempting a polynomial construction of degree at least 2, with coefficients yet to be determined.
 - (C4) **3P:** Constructions that fail for finitely many cases. (e.g. $b = n^2 - n$ fails for $n = 1$)
 - (C5) **0P:** Constructions that fail for infinitely many cases.
- (D) **0P:** Investigating the case where $a = b$ or one of a and b is fixed.

Items C1, C2, C3 are additive up to a maximum of 2 points. No points are deducted for not arguing that the expression is positive.

N2) Determine all triples (a, b, p) of positive integers where p is a prime number and

$$p(a + b) = a^2(2p^2 - pb + 1).$$

Solution

Answer: All triples (a, b, p) that satisfy the equation are of the form $(2p, 2p, p)$ for any prime p .

Solution: Since p divides the left hand side of the equation it must also divide the right hand side. Notice that $\gcd(p, (2p^2 - pb + 1)) = 1$ therefore $p|a^2$. Since p is prime and $p|a^2$ we have $p|a$. By substituting $pk = a$ into the equation and dividing by p on both sides we get

$$pk + b = pk^2(2p^2 - pb + 1)$$

Now pk divides the right hand side and therefore it must also divide $pk + b$. Since $pk|pk$ we get that $pk|b$. Again by substituting $pk l = b$ into the equation and dividing by pk on both sides we get

$$1 + l = k(2p^2 - klp^2 + 1)$$

The left hand side $1 + l$ is clearly positive and therefore the right hand side must be positive as well. Since k is positive $(2p^2 - klp^2 + 1)$ must be positive as well. Now notice that if $kl \geq 3$ then $2p^2 - klp^2 + 1 \leq -p^2 + 1 \leq -4 + 1 < 0$ since the smallest prime is 2. Therefore $kl \leq 2$ and there are three cases to check.

Case 1: If $k = 1$ and $l = 1$ then $2 = p^2 + 1$ which is clearly impossible.

Case 2: If $k = 1$ and $l = 2$ then $3 = 1$ which again is clearly impossible.

Case 3: If $k = 2$ and $l = 1$ then $2 = 2$ which is true for any prime p .

Since $pk = a$ and $pk l = b$ gives a valid solution only for any p where $k = 2$ and $l = 1$, all triples (a, b, p) which are solutions must be of the form $(2p, 2p, p)$. This can be checked by plugging in $a = 2p$ and $b = 2p$ into our original equation.

Marking Scheme

- (A) **1P:** Verify that $(2p, 2p, p)$ works for any prime p .
- (B) **3P:** Show $p|b$
 - (B1) **1P:** Show $p|a^2$
 - (B2) **1P:** Show $p|a$ and write $a = pk$ (or consider $\frac{a}{p}$)
- (C) **2P:** Show $b \leq 2p$
 - (C1) **1P:** State that $(2p^2 - pb + 1)$ must be positive
- (D) **1P:** Case analyse $kl \leq 2$ properly.

The marking scheme is additive.